

muRelBench

MicroBenchmarking for Zonotope Domains

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Outline

① Background

Weakly-Relational Abstract Domains

② Motivation

③ Framework

④ Experimental Results

⑤ Conclusions

Weakly-Relational Abstract Domains

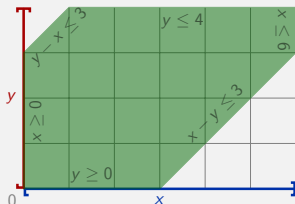


Figure: Projection of Zones

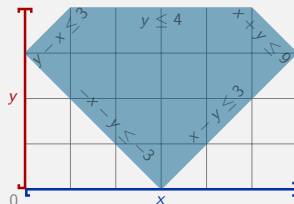


Figure: Projection of Octagons

Definition

Weakly-Relational Domains: n -dimensional conjunction of linear inequalities, representing a sub-class of Polytopes.

- Must be convex.
- Restricted to integer, linear inequalities.
- Restricted to unit two variables per inequality (uTVPI).

Octagons Abstract Domain

$$x + y \leq 3$$

$$x - y \leq 4$$

$$y - x \leq 3$$

$$y \leq 3$$

$$y \geq 1$$

Octagons

Encode unit sums between two program variables, also known as unit two variables per inequality (uTVPI) ^a.

^aMiné, 2006, “The Octagon Abstract Domain”, Gange et al., 2021, “A Fresh Look at Zones and Octagons”

Octagons Abstract Domain

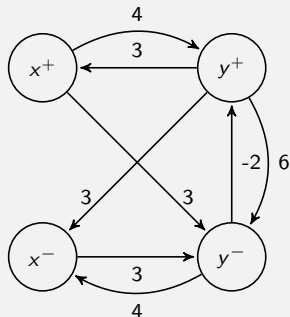
$$x + y \leq 3$$

$$x - y \leq 4$$

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Octagons Abstract Domain

Transitive Closure

$$x + y \leq 3$$

$$y - x \leq 3$$

$$y \leq 3$$

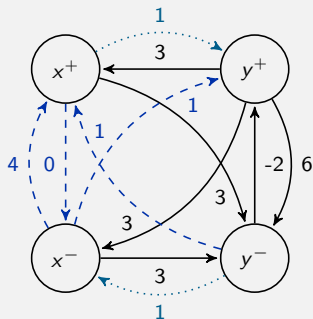
$$y \geq 1$$

$$x \leq 0$$

$$x \geq -2$$

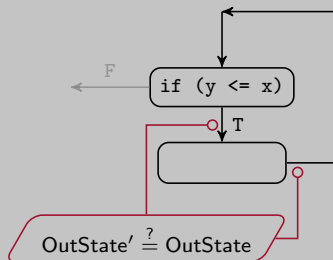
$$x - y \leq 1$$

$$-x - y \leq 1$$



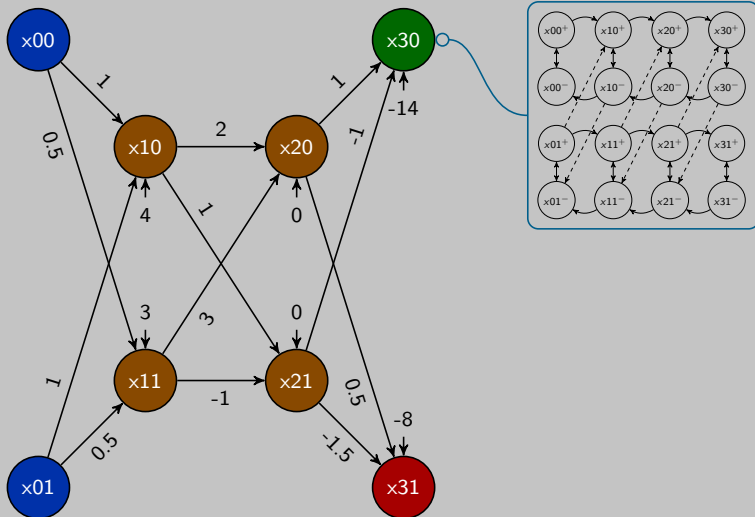
Transitive closure is repeatedly called during analysis

Data-flow analysis of Programs



Transitive closure is repeatedly called during analysis

Analysis of Neural Networks



The Problem

How do you evaluate new developments?

You have developed a new closure implementation for an Octagons

Evaluate on small programs

Closure	Program	Mean (ms)	σ
Floyd-Warshall	Fibonacci	144	32.2
	Loop	46.8	3.1
Chawdhary	Fibonacci	117	5.1
	Loop	49.6	10.3

Table: Small programs used to demonstrate performance characteristics of using different closure algorithms.

muRelBench attempts to provide the following:

Goals

- A framework for benchmarking weakly-relational domain operations
- Generates (Feasible) Synthetic Instances of the target domains
- Provides flexible opportunities for extension
- Supports oracles to verify correctness of operations

Framework Component Overview

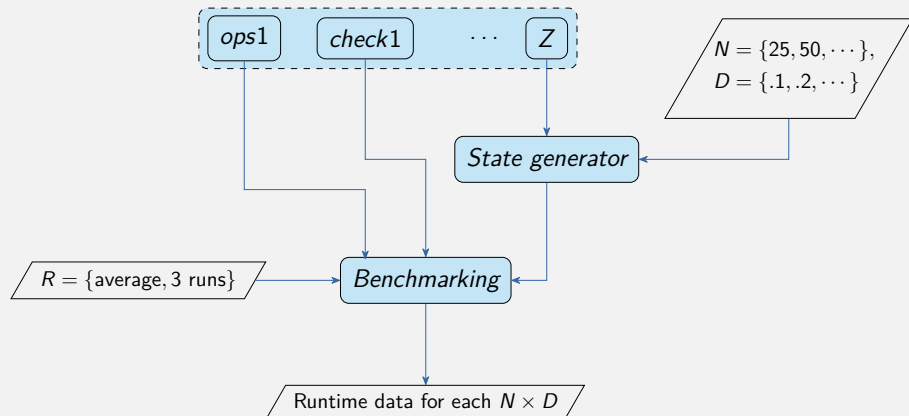


Figure: Component diagram of `muRelBench`, specifying the framework's and user-defined components.

Comparison of Full-Closure to Incremental

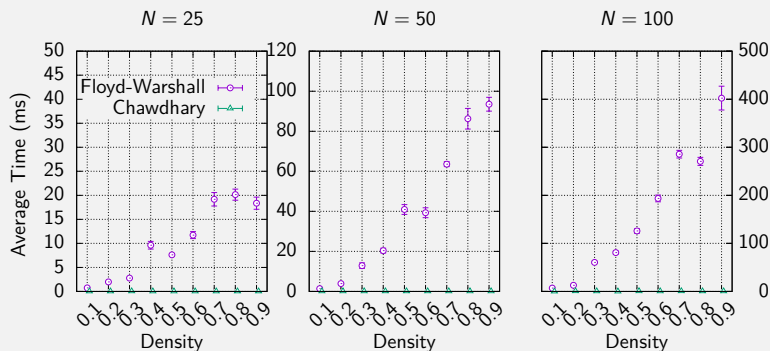


Figure: Plots of microbenchmark results of closure operations, each subplot varies the number of variables, each sample varies the connectivity of program variables.

Incremental compared to Code-Motion Incremental

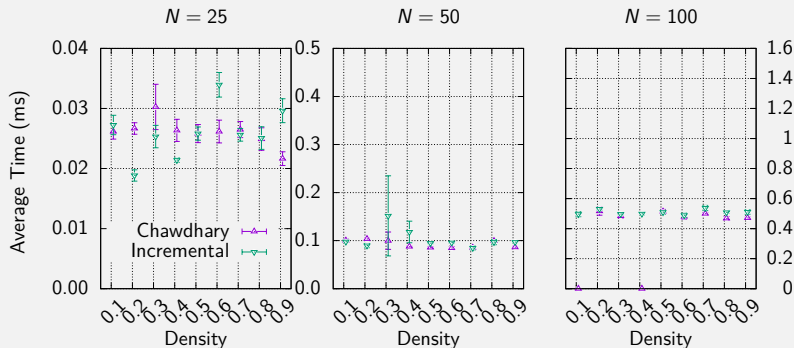


Figure: Plots of microbenchmark results for incremental closure operations. Each subplot again varies by number of variables, each sample varies by connectivity of the program variables.

Conclusions and Future Work

muRelBench provides:

- a framework for benchmarking operations for Zonotopes
- an extensible framework for additional abstract domains
- automatic report generation via GitHub actions
- optional oracles for verification

Conclusions and Future Work

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We invite contributors to the following areas:

- Additional abstract domains
- Additional operation implementations

Thank you

Code on
GitHub



fmsea/muRelBench

Questions?

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References I

- [1] Graeme Gange et al. “A Fresh Look at Zones and Octagons”. In: *ACM Transactions on Programming Languages and Systems* 43.3 (Sept. 2021), pp. 1–51. ISSN: 1558-4593. DOI: 10.1145/3457885. URL: <http://dx.doi.org/10.1145/3457885>.
- [2] Antoine Miné. “The Octagon Abstract Domain”. In: *Higher-Order and Symbolic Computation* 19.1 (Mar. 2006), pp. 31–100. ISSN: 1573-0557. DOI: 10.1007/s10990-006-8609-1. URL: <http://dx.doi.org/10.1007/s10990-006-8609-1>.

Encoding for Octagons

Program Variable Space

$$\begin{aligned}
 x + y &\leq c \\
 x - y &\leq c \\
 y - x &\leq c \\
 -x - y &\leq c \\
 2y &\leq c \\
 2y &\geq c
 \end{aligned}$$

 $\rightsquigarrow$

Octagon Variable Space

$$\begin{aligned}
 x^+ - y^- &\leq c \wedge y^+ - x^- \leq c \\
 y^+ - x^+ &\leq c \wedge x^- - y^- \leq c \\
 x^+ - y^+ &\leq c \wedge y^- - x^- \leq c \\
 x^- - y^+ &\leq c \wedge y^- - x^+ \leq c \\
 y^+ - y^- &\leq 2c \\
 y^- - y^+ &\leq 2c
 \end{aligned}$$